ECE 2564
Fall 2025
Homework 3

Please sign (You can type your name in place of signature.)

"I have neither given nor received unauthorized assistance on this assignment."

To complete this homework, you need to complete the following steps:

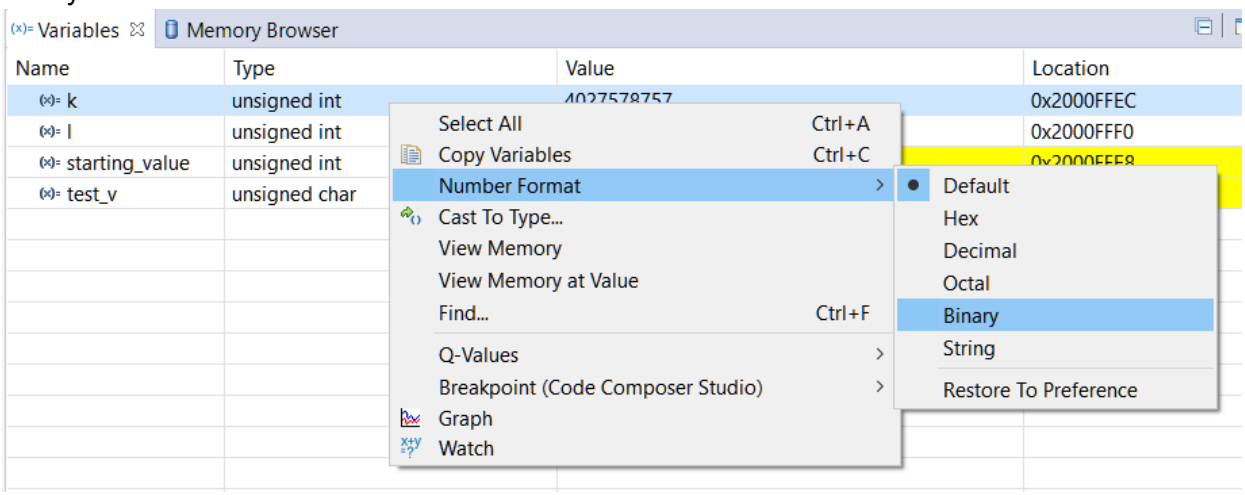
1. Sign the Honors Code statement.
2. Accept the following GitHub classroom assignment: https://classroom.github.com/a/r3wzhCvY
3. Import your homework repository to CCS.
4. There are two projects for this homework. Follow the instructions below to complete each problem. As you follow the instructions, you are developing code in CCS and creating an answer/solution document. The solution document could be this Word document where you have entered your answers, screenshots, etc. Alternatively, you can create a document from scratch.
5. Once you have finished each problem, submit your code to GitHub.
6. Turn your solution document (perhaps a modified version of this document) to Canvas.
7. You will be graded based on the code you submit on GitHub and the document you submit to Canvas.
8. All the C header files required for this homework are already included in the starter kit. You are not allowed to add/include any additional header files to your source code.

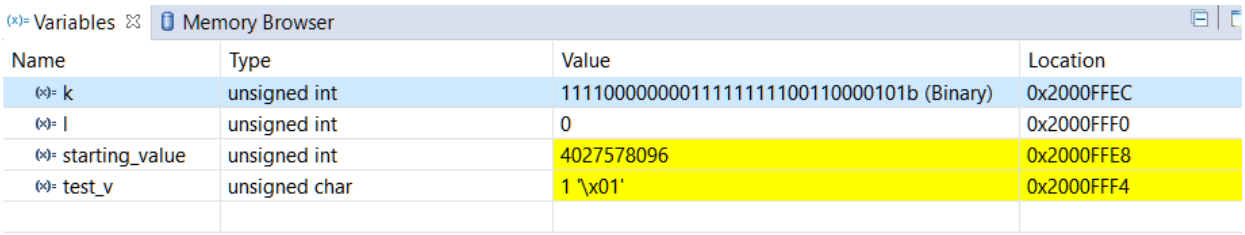
Problem 0. Enter your GitHub repository link here.

Problem 1. (25 points)

In this problem, you will experiment with setting, resetting and toggling bits of a given integer. **You are required to use predefined macros, BIT0, BIT1, BIT2, … or define your own bitmasks to achieve this.** Open the project related to problem 1 and read the whole code including comments. There are to-do items in the code that you have to complete. After finishing each to-do, you are welcome to compile and debug the code to make sure your code works as you expect it.

Tip: You can view variables in binary by right clicking on variable row > Number Format > Binary:





A. (20 points) Copy-paste your whole code from hw3_p1_main.c to your solution document.

B. (5 points) Show a screenshot of the debugger window at the end of your debugging session. Make sure the Variables view is open. The **starting_value should be something you have chosen**. Also, since you have written code for updating k, k should be different from starting_value and your test_value could be true or false depending on your k.

Problem 2. (75 points)

In this problem, you will experiment with structures, pointers, macros, enum, typedef and bit manipulation. Consider a stage lighting system. Each light in the system has an x and y coordinate that reflects its location on the stage which can be between 0 and 100. Each light also has a register that determines its status, called Light Status Register or LSR. For each LSR, the bits control the light based on the below specification.

- Bit 0: controls whether the light flickers: 0 no flicker, 1 flicker.
- Bit 2-1: control the light intensity: 00 off, 01 low, 10 medium, 11 high.
- Bit 4-3: control the color: 00 white, 01 blue, 10 red, 11 green.
- Bit 7-5: unused.

Examples:

- If we write 0x00 to an LSR, it means that the corresponding light should be off.
- If we write 0xFF to an LSR, it means that the corresponding light should be green, high intensity, and flickering.

Our goal is to develop a project that can control this lighting system with an **easy-to-read** and **easy-to-debug** code.

A. (3 points) Write the value of an LSR that programs a light with red color, low intensity and no flicker. Show the value both in binary and hex.

B. (3 points) Translate what it means if the value of an LSR is 0x9A.

C. (14 points) Open the project associated with problem 2 in CCS and study the whole code, **especially the comments**! Complete TODOs #1 through #5. These to-do items only set the stage for developing code. Copy-paste your code for to-do items 1 through 5. ***Once you finish this step, your code should compile without errors.***

D. (8 points) Complete to-do item #6, which is developing three functions and debugging them. (The first function is given to you as a guide.) Copy-paste the solution to your three functions below. (The placeholder for these functions is below the main function.)

E. (2 points) Debug your code for item #6 and take a screenshot and put it here. Change tempLSR to 0x9A before you debug. You can check your code against the answer you provided to part B. They should match! Notice how in the variable view the enum variables are shown with their enum instance. For example, the variable flickering will be either FLICK_ON or FLICK_OFF in the debugger after you are done debugging this section.

F. (18 points) Complete to-do item #7 by developing another three functions. Copy-paste the solution to your three functions below. (The placeholder for these functions is below the main function.)

G. (2 points) Debug the lines of code related to item #7 after updating them to produce your answer in part A. Take a screenshot of your debugger output. What is the value of tempLSR? Check it against your answer to part A; they should match!

H. (8 points) Complete item #8 and copy-paste the solution for the code you have written for makeLSR. (The placeholder for this function is below the main function.)

I. (2 points) Debug the lines that test makeLSR. Take a screenshot of the debugger window and paste it here.

J. (18 points) Write this function increaseLight for item #9. To receive full points, you need to call these functions within this function:
   1) getIntensitySetting
   2) updateIntensity

   Copy-paste your answer to increaseLight function here. (The placeholder for this function is below the main function.)

K. (2 points) Debug item #9, can you confirm that the light intensity of myLSR has increased?

L. (18 points) Complete to-do item #10. This is the first time you are writing any code inside the main function. You need to call the functions you developed in the earlier steps to achieve your goal. Copy-paste your snippet of code here.

M. (2 points) Debug this last part. Make sure the lights with the x and y coordinates that fall under the conditions described in the to-do item are updated correctly. Take a screenshot of the debugger window and attach it here.